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We evaluate the vibrational theory of olfaction, which posits that insects can detect infrared electromagnetic radiation emitted by semiochemicals alongside or in addition to molecular binding mechanisms. Using morphological measurements, spectroscopic analyses, neural latency comparisons, and tested computational models implemented in `src/`, we assess the plausibility and testable predictions of this hypothesis.

Key results: (i) Sensilla geometry aligns with predicted resonant wavelengths derived from dielectric waveguide models; (ii) published olfactory receptor neuron latencies (1–5 ms) are consistent with rapid, non-diffusion-limited detection; (iii) cuticular hydrocarbon (CHC) spectral peaks fall within modeled atmospheric transmission windows (2–5 m, 8–14 m, 17–25 m) using `src/core.calculate_atmospheric_transmission` (validated in `tests/test_core.py`).

Framework: We integrate resonant cavity and waveguide theory with CHC spectroscopy (`src/spectroscopy.py`) and sensilla morphology (`src/sensilla.py`). Implementations are covered by targeted unit tests (e.g., `tests/test_core.py::TestAtmosphericTransmission`, `tests/test_sensilla.py::TestSensillaAnalysis`, `tests/test_spectroscopy_analysis.py::TestAnalyzeChcSpectra`).

Scope and implications: Findings provide falsifiable predictions for wavelength tuning, behavioral IR-only responses, and neural responses to IR stimulation. If validated, these mechanisms have implications for sensory biology and biomimetic sensing.

Reproducibility: Analyses are deterministic (fixed seeds) and reproducible via the repository’s unified pipeline; see the reproducibility subsection in Section ?? and the functions mapped to tests in ?? and the Symbols/Glossary.